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MAKERERE UNIVERSITY BUSINESS SCHOOL

FINAL EXAMINATION FOR THE DEGREE OF
BACHELOR OF BUSINESS COMPUTING
OF MAKERERE UNIVERSITY ACADEMIC YEAR 2018/2019

COURSE NAME: SYSTEMS ANALYSIS AND DESIGN. YEAR OF STUDY: 2

COURSE CODE: BUC 2101

DATE: 29th November 2018

SEMESTER: 1

TIME: 2:00 PM - 5:00 PM

Answer any FOUR questions

Question One:

The Systems Development Life Cycle has a similar set of four fundamental phases: planning, and implementation. Different projects may emphasize different parts of the SDLC or apply phases in different ways, but all projects have elements of these four phases. Each phase is its series of steps, which rely upon techniques that produce deliverables.

Discuss the following phases of systems development life cycle

- a) Planning (7 marks)
- b) Analysis (6 marks)
- c) Design (6 marks)
- d) Implementation (6 marks)

Question Two:

Human beings influence input design. Inputs should be as simple as possible and design possibility of incorrect data being entered. The needs of system users must be considered. Several human factors should be evaluated. The volume of data to be input should be minimized. That is input, the greater the potential number of input errors and the longer it takes to input the

- a) Explain any five general principles that should be followed when designing forms (1 marks)
- b) Explain human engineering factors that should be incorporated in the input design (9 marks)
- c) Explain three levels of software requirements (6 marks)

Question Three:

Today, it is hard to imagine any industry or business that has not been affected by computer system and computer applications. Many businesses consider management of their information equal in importance to managing their other key resources: Property, facilities, equipment capital. Information system development projects do not succeed by chance, successful information system development projects follow a systematic "Analysis and Design" Process.

- a) Identify any five stakeholders in information systems development project and explain the role of each in systems development (10 marks)
- b) Explain any six qualities of a modern systems analyst (12 marks)
- c) Explain the term "Problem" in three different perspectives in information systems development (6 marks)

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Question Four:

- a) The following concepts are commonly used in security and cryptography, explain between them.
- i. Symmetric key algorithms and asymmetric key algorithms (3 marks)
 - ii. Substitution and transposition techniques (3 marks)
 - iii. Password and biometric based security system (3 marks)
- b) The acquisition and sharing of secure information has been enhanced with the use of the Internet. The transmission of secured documents/communication is usually indicated with the URL <https://www....> With an illustration, explain how this information is secured while being transmitted over the Internet. (6 marks)
- c) The most important component of any developed system is the output interface. Discuss the importance of the output interface in the designing of the output interface of a system (10 marks)

Question Five:

- a) Using an illustration, explain the role played by the primary key and foreign key in the development of a relational database. (5 marks)
- b) The V-model depicts quality assurance in system development. With an illustration of the V-model, discuss the various tests that address quality assurance in system development. (10 marks)
- c) With illustrations explain the following: (2 marks each)
- i. Truman's Triangle
 - ii. Gantt chart
 - iii. Network diagram
 - iv. Project Management Triangle (Triple constraint)
 - v. Project Management Diamond (Quadruple constraint)

Question Six:

- a) Explain the difference between file and database, and provide an example from the Microsoft applications that can represent each one of them (3 marks)
- b) There are various traditional and modern techniques/methods of collecting data/information for system requirement determination. Describe five (5) traditional and one (1) modern technique

Question Six:

- a) Explain the difference between file and database, and provide an example from the Microsoft applications that can represent each one of them (3 marks)
- b) There are various traditional and modern techniques/methods of collecting data/information c system requirement determination. Describe five (5) traditional and one (1) modern technique that can be used in system requirement determination. (12 marks)
- c) There are different symbols used in data flow diagrams (DFD) to depict the various processes system. Using these symbols, apply them in a dataflow diagram of the MUBS library system decomposed to level 2 (10 marks)